

HOSSAM ABDELHAKIM ORABY

SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

Site Reliability Engineer with 4+ years of experience architecting scalable systems, specializing in C++, Python, and CI/CD automation. Proven track record in building high-performance systems, cloud deployments, and troubleshooting complex distributed environments. Microsoft Certified, with strong experience in Azure, Linux administration, and service reliability. Passionate about leading teams, optimizing performance, and delivering robust solutions for mission-critical applications.

SKILLS

Languages: C++ (C++11/14/17), Python, Go, JavaScript (Node.js)
Tools & Platforms: Linux, Kubernetes, Azure DevOps, Docker, Jenkins, Git, Terraform, FastAPI.
Databases: MongoDB, MySQL, SQLite, Redis, Elasticsearch.
AI/ML: Built LLM-based pipelines (LangChain, RAG) and integrated AI into production systems.
Concepts: Multithreading, Distributed Systems, Performance Optimization, RESTful APIs, Microservices, Cloud Infrastructure, CI/CD Pipelines
Problem Solving: Advanced Algorithms, Data Structures, Competitive Programming.

PROFESSIONAL EXPERIENCE

C++ Software Engineer at Tahaluf AI Emarat - Abu Dhabi, UAE **Apr 2023 – Dec 2024**

- Managed high-performance C++ systems as part of AI R&D, focusing on smart city and traffic management solutions.
- Refactored legacy C++ repositories, improving memory usage and maintainability by over 25%.
- Implemented advanced caching and multithreading techniques, increasing system responsiveness by 30%.
- Managed multi-repository builds with CMake, Docker, reducing build errors and deployment times.
- Led debugging and performance optimization of distributed systems processing real-time data streams.
- Designed orchestration layers for distributed subsystems, ensuring scalability and resilience.

Software Engineer / Intelligent Systems Developer at Upwork **Mar 2023 – Dec 2024**

- Delivered 11+ projects across US, EU, and MENA markets, specializing in C++ and Python automation tools.
- Built production-ready solutions using FastAPI, Docker, Kubernetes, and microservice architectures.
- Automated workflows via Python scripts, reducing development cycles by 75%.
- Managed cloud deployments and complex troubleshooting in distributed systems.
- Maintained 100% on-time delivery and top client ratings.

C++ Software Engineer at Medsoft - Cairo, Egypt **Jan 2023 – Apr 2023**

- Created surgical simulation tools in C++, improving training accuracy and planning time by 50%.
- Developed modules deployed in US hospitals for bone transplant simulations.
- Applied algorithmic optimizations to reduce computational time in medical simulations.

Backend Software Engineer at Fatura - Cairo, Egypt **Sep 2021 – Dec 2024**

- Developed RESTful APIs using Node.js, connecting over 200 suppliers and distributors.
- Optimized MongoDB queries, achieving 55% faster database performance.
- Contributed to the scalability of B2B systems, supporting rapid company growth.

PROJECTS

WikiDocify – Collaborative Documentation System	2025
• Architected a platform using Go, Kubernetes, Docker, Elasticsearch, PostgreSQL. • Integrated LLMs for AI-driven document summarization and search. • Deployed scalable services with automated CI/CD pipelines.	
mini-rag – Retrieval-Augmented Generation API	2025
• Developed backend in Python FastAPI, deployed via Docker and Kubernetes. • Implemented document ingestion and chunking pipelines for NLP tasks.	
Real-Time Chat Application	2021
• Built scalable real-time chat platform using Node.js, WebSocket, Docker, and Jenkins CI/CD. • Managed 1,000+ concurrent users with sub-100ms message latency.	
Arabic Language Compiler	2020
• Developed a compiler in C# for Arabic linguistic processing, improving parsing efficiency. • Created custom lexical analyzers and syntax parsing algorithms.	

EDUCATION

Bachelor of Computer Science	2017 - 2021
Faculty of Computers and Information, Assiut University, Egypt	

VOLUNTEER AND EXTRACURRICULAR ACTIVITIES

• Founder & Lead Instructor ABCL Academy	Jun 2024 – Present
◦ Founded educational initiative teaching advanced algorithms, data structures, and C++. ◦ Mentored 3,000+ students and coached 7 teams to ICPC, ECPC, and ACPC finals. ◦ Delivered 25+ workshops at universities, promoting competitive programming.	
• Coach & Judge Egyptian Olympiad in Informatics (EOI)	
◦ Founded ICPC Communities (EELU Qena, SVU) - Coached 7 university teams to ECPC/ACPC finals; founded communities. ◦ Mentored school students, achieving 1 Gold, 2 Silver, 5 Bronze medals (7th nationally)	
• YouTube Channel Founder, MaTb3a:	
◦ Produced over 400 videos on C++, algorithms, and FAANG interview prep. ◦ Grew channel to 7.8K subscribers, serving Arabic-speaking software engineers.	

CERTIFICATES

- Microsoft Certified: Azure AI Fundamentals (2021)
- DevOps Mastery Specialization – KodeKloud (2024)
- Certified Problem Solving Coach – Egyptian Olympiad in Informatics (EOI)
- ALX AiCE – AI Career Essentials Graduate (2024)

HONORS & AWARDS

- Top 0.4% Global Ranking on LeetCode (1,100+ day streak)
- ACM Arab Collegiate Programming Contest (ACPC) Finalist – 46th Place (2021)
- Upper Egypt Collegiate Programming Contest (ECPC) – 2nd Place (2020)

SOFT SKILLS

- Leadership and Mentoring
- Excellent Communication
- Agile Development
- Problem-Solving under Pressure